

CMPE 492

Skill Learning in Robotic Manipulation

Serdar Bahar

Advisor:

Assoc. Prof. Emre Uğur

TABLE OF CONTENTS

1. INTRODUCTION	1
1.1. Broad Impact	1
1.2. Ethical Considerations	2
2. PROJECT DEFINITION AND PLANNING	3
2.1. Project Definition	3
2.1.1. Problem Statement	3
2.1.2. Objectives	3
2.1.3. Scope	4
2.1.4. Contributions	4
2.2. Project Planning	4
2.2.1. Project Time and Resource Estimation	4
2.2.2. Success Criteria	5
2.2.3. Risk Analysis	5
3. RELATED WORK	7
3.1. Related Work	7
3.1.1. Movement Primitives and Policy Representations	7
3.1.2. Generalization in Policy Learning	7
3.1.3. Diffeomorphisms and Space Transformations	8
4. METHODOLOGY	9
5. IMPLEMENTATION AND TESTING	10
5.1. Implementation	10
6. RESULTS	11
6.0.1. Preliminary Experiment: Unconstrained Optimization vs. Kernel- Based Prediction	11
7. CONCLUSION	13
REFERENCES	14

1. INTRODUCTION

1.1. Broad Impact

Learning from demonstrations (LfD) enables robots to acquire complex manipulation skills without requiring explicit programming of possible scenarios. However, a fundamental limitation of current approaches is their inability to generalize beyond the distribution of demonstrated conditions. When a robot encounters a novel situation, such as a previously unseen object pose, an unfamiliar joint configuration, or a slightly different task configuration, policies trained on limited demonstrations often produce degenerate or unsafe outputs. This brittleness is a major barrier to deploying learned robotic manipulation skills in unstructured, real-world environments such as households, hospitals, and manufacturing floors.

This project aims to address this challenge by investigating learned spatial transformations that convert extrapolation queries into interpolation queries within the space of demonstrations. If successful, the proposed approach would significantly expand the operational strength of learned manipulation policies, reducing the number of demonstrations required for reliable deployment. Fewer required demonstrations also translates directly to reduced setup time, lower cost, and broader accessibility of robotic manipulation systems.

More generalizable robotic manipulation has the potential to accelerate the adoption of assistive robots in elder care and rehabilitation, where the diversity of environments and user needs makes exhaustive demonstration collection impractical. In manufacturing, improved generalization would enable more flexible automation pipelines that can adapt to product variations without costly reprogramming. Furthermore, the mathematical framework explored in this work, learning diffeomorphisms and space-warping transformations for interpolation, is not limited to robotics; it has potential applications in other domains where function approximation from sparse data is criti-

cal.

1.2. Ethical Considerations

The experiments in this project will involve simulated environments and do not include human subjects, personal data, or safety-critical applications.

2. PROJECT DEFINITION AND PLANNING

2.1. Project Definition

2.1.1. Problem Statement

Learning from demonstrations (LfD) enables robots to acquire manipulation skills by observing a set of expert trajectories. A core challenge arises when the robot must execute the learned skill under conditions not represented in the demonstration set (novel object positions, unseen joint configurations, or different task parameters). In such cases, the policy must *extrapolate* beyond the convex hull of training conditions, which typically leads to unreliable or degenerate outputs. This project investigates how learned nonlinear transformations of the robot’s configuration space can convert extrapolation queries into interpolation queries, thereby enabling reliable generalization from limited demonstrations.

2.1.2. Objectives

The primary objective of this project is to develop and evaluate a framework for learning transformations that reshape the configuration space of a robot so that novel queries fall within the convex hull of the transformed demonstration set. The specific objectives are:

- (i) **Differentiable convex hull membership:** Develop a differentiable surrogate for testing whether a point lies inside the convex hull of a given point set, to use it as a training signal for learning transformations.
- (ii) **Learned space transformations:** Train a neural network to learn a nonlinear mapping of the input space such that, after transformation, previously out-of-hull queries become in-hull, effectively converting extrapolation into interpolation.
- (iii) **Proof-of-concept validation:** Validate the approach on synthetic point datasets

in varying dimensionalities, comparing the differentiable convex hull surrogate against baseline methods.

- (iv) **Application to robotic manipulation:** Apply the learned transformation framework to a robot policy setting, where the condition space corresponds to task parameters and the policy outputs joint-space trajectories.

2.1.3. Scope

The current scope of this project focuses on the mathematical and algorithmic foundations: formulating the differentiable convex hull problem, and learning space transformations on synthetic data. The extension to real or simulated robotic manipulation tasks with trajectory-level policies (e.g., movement primitives) is planned for the final phase of the project. The project targets the algorithmic contribution of learned spatial transformations for generalization.

2.1.4. Contributions

The expected contributions of this project is to provide a complete methodology for learning transformations that demonstrably changes the configuration space of robot such that, out-of-distribution (extrapolating) points become in-distribution (interpolating).

2.2. Project Planning

2.2.1. Project Time and Resource Estimation

Phase 1 - Literature Review and Theoretical Framework (Weeks 1-6): The phase involves a review of the literature on learning from demonstrations, movement primitives (DMPs, ProMPs, ProDMPs), policy generalization, and extrapolation in machine learning. Key mathematical concepts including diffeomorphisms, normalizing flows, convex hull geometry, and kernel methods are to be investigated.

Phase 2 - Implementation and Algorithm Development (Weeks 7-11):

This phase involves the implementation and the algorithmic development of the proposed transformation learning. Implementation side should follow good programming practices. The developed algorithms should be tested continually with synthetic trajectories. Modular classes and effective visualizations of the learned mappings are expected to facilitate this process.

Phase 3 - Experimentation and Evaluation (Weeks 12-13): The phase focuses on completing the learned transformation experiments, validating that the MLP-based mapping successfully expands convex hull coverage for out-of-distribution points, and preparing the final report and poster. If time permits, an initial application to trajectory-level robot policy generalization will be explored.

2.2.2. Success Criteria

- The learned MLP transformation maps out-of-hull points closer to or inside the convex hull of the transformed demonstration set, as measured by reduced reconstruction scores for Class 1 points after transformation.
- The overall framework is validated on synthetic data in dimensionalities representative of robot policy condition spaces (e.g., 12 dimensions).

2.2.3. Risk Analysis

Several risks were identified during the planning and execution of this project:

- **Scalability of differentiable convex hull surrogates:** As the dimensionality of the space increases, the convex hull becomes increasingly sparse and the distinction between in-hull and out-of-hull points may become less meaningful.
- **Training instability of the learned transformation:** The MLP-based transformation must be trained with the differentiable convex hull surrogate as part

of the loss function.

- **Collapse of the learned mapping:** The transformation network could learn a trivial solution (e.g., mapping all points to a single location) that minimizes the convex hull objective without producing a useful transformation. Regularization strategies such as bijectivity constraints penalties may be needed.
- **Gap between synthetic and robotic settings:** The current validation uses synthetic point data, and the transfer to actual trajectory-level policy generalization may reveal additional challenges. This risk is accepted for the scope and will be attempted to be addressed in the final phase.

3. RELATED WORK

3.1. Related Work

The ability to generalize learned behaviors to novel conditions is a central challenge in robot learning from demonstrations. This section reviews the relevant literature on policy representations, generalization and extrapolation strategies, and the mathematical tools that is related to this project.

3.1.1. Movement Primitives and Policy Representations

Movement primitives provide a compact, parameterized representation of robot trajectories. Dynamic Movement Primitives (DMPs) [1] encode trajectories as dynamical systems with learnable forcing terms, enabling temporal scaling and goal adaptation. Probabilistic Movement Primitives (ProMPs) [2] extend this to a probabilistic framework by maintaining distributions over trajectory weights, allowing conditioning on via-points and blending of multiple demonstrations. More recently, ProDMPs [3] combine the strengths of both formulations. While these representations support a degree of generalization through conditioning and interpolation within the demonstrated set, their performance degrades when queried at conditions far from the training distribution.

3.1.2. Generalization in Policy Learning

Several lines of work address the generalization of learned policies beyond demonstrated conditions. Task-parameterized models [4] achieve generalization by learning policies in multiple local reference frames and combining them according to task parameters, but remain limited to the span of observed frames. Diffusion-based policy models [5] and flow-matching approaches have shown strong interpolation capabilities by learning to denoise trajectories conditioned on scene observations, yet they do not

explicitly address extrapolation. Foundation models for robotic manipulation [6] pursue generalization through massive-scale data collection and transfer learning, which sidesteps extrapolation by attempting to cover the full distribution. In contrast, this project takes a complementary approach: rather than expanding the dataset, it transforms the space so that extrapolation reduces to interpolation.

3.1.3. Diffeomorphisms and Space Transformations

Diffeomorphisms (smooth, invertible mappings with smooth inverses) have been used in various contexts to reshape input spaces. Normalizing flows [7, 8] learn diffeomorphic transformations for density estimation by mapping complex distributions to simple base distributions. In the context of dynamical systems, diffeomorphisms have been applied to transform stable policies into topologically equivalent ones operating in different coordinate frames [9]. The key idea in this project is related: learning a diffeomorphic mapping that warps the condition space so that the convex hull of the transformed demonstrations encloses a larger region.

4. METHODOLOGY

5. IMPLEMENTATION AND TESTING

5.1. Implementation

The project is implemented in Python using PyTorch as the primary deep learning framework, with the codebase managed through `uv` for reproducible dependency resolution. The source code is organized as an installable package under `src/`, with experiment scripts in `scripts/`. The synthetic dataset (`CustomPointDataset`) generates labeled point sets with controllable convex hull membership by sampling points both inside and outside the convex hull of a base point set, parameterized by an epsilon margin and a configurable sampling distribution across in-hull, boundary, and out-of-hull regions. The unconstrained optimization surrogate runs a 50-step Adam inner loop over softmax-parameterized convex weights to minimize reconstruction error for each query, while the kernel-based surrogate computes a closed-form distance-weighted softmax average, avoiding any iterative optimization. Random projection matrices are used to evaluate the methods in higher-dimensional spaces (e.g., 12 dimensions) from the base low-dimensional dataset. The code for the project is available at https://github.com/serdarbahar/skill_learning_in_robotic_manipulation.

6. RESULTS

6.0.1. Preliminary Experiment: Unconstrained Optimization vs. Kernel-Based Prediction

To validate the differentiable convex hull surrogates, an experiment was conducted comparing the two proposed methods: unconstrained optimization and kernel-based prediction on synthetic data projected to 12 dimensions. The dataset consists of 1000 points with binary labels: Class 0 (in-hull) and Class 1 (out-of-hull). For each data point, both methods attempt to reconstruct it as a convex combination of a set of test points (vertices), and the reconstruction error serves as the membership score. Lower scores indicate that the point is well-approximated by a convex combination of the vertices, suggesting membership in or proximity to their convex hull. The experiment was repeated with varying numbers of test points (2, 4, 8, 12, 16, and 20) to assess robustness.

Results. Table 6.1 summarizes the results. Both methods show the desired behavior: Class 0 (in-hull) points consistently receive higher reconstruction scores than Class 1 (out-of-hull) points under unconstrained optimization, indicating that unconstrained optimization achieves near-zero error for in-hull points. Specifically, the unconstrained optimization method yields Class 1 scores near zero across all test point counts (e.g., 0.0158 ± 0.0239 with 2 test points, 0.0028 ± 0.0048 with 8 test points), confirming that it can almost perfectly reconstruct out-of-hull points as convex combinations, which is expected since the optimization is unconstrained in the softmax space and can exploit the geometry.

Table 6.1. Comparison of unconstrained optimization and kernel-based prediction across varying numbers of test points ($d = 12$, $N = 1000$).

Test Points	Overall Score		Class 0 (Out-Hull)		Class 1 (In-Hull)		Time (s)	
	Unconstr.	Kernel	Unconstr.	Kernel	Unconstr.	Kernel	Unconstr.	Kernel
2	1.27 ± 1.08	1.82 ± 1.94	1.58 ± 0.99	1.35 ± 0.94	0.02 ± 0.02	3.71 ± 3.32	0.0030	≈ 0
4	0.71 ± 0.59	0.67 ± 0.52	0.88 ± 0.53	0.77 ± 0.50	0.00 ± 0.01	0.27 ± 0.35	0.0026	≈ 0
8	0.69 ± 0.57	0.67 ± 0.54	0.86 ± 0.51	0.82 ± 0.50	0.00 ± 0.00	0.06 ± 0.08	0.0026	≈ 0
12	0.83 ± 0.64	0.85 ± 0.58	1.03 ± 0.55	1.04 ± 0.50	0.01 ± 0.01	0.11 ± 0.12	0.0026	≈ 0
16	0.68 ± 0.52	0.75 ± 0.51	0.85 ± 0.44	0.92 ± 0.42	0.00 ± 0.01	0.06 ± 0.06	0.0026	≈ 0
20	0.96 ± 0.73	0.95 ± 0.67	1.20 ± 0.62	1.17 ± 0.56	0.01 ± 0.02	0.05 ± 0.05	0.0027	≈ 0

7. CONCLUSION

REFERENCES

1. Ijspeert, A. J., J. Nakanishi, H. Hoffmann, P. Pastor and S. Schaal, “Dynamical Movement Primitives: Learning Attractor Models for Motor Behaviors”, *Neural Computation*, Vol. 25, No. 2, pp. 328–373, 2013.
2. Paraschos, A., C. Daniel, J. R. Peters and G. Neumann, “Probabilistic Movement Primitives”, C. Burges, L. Bottou, M. Welling, Z. Ghahramani and K. Weinberger (Editors), *Advances in Neural Information Processing Systems*, Vol. 26, Curran Associates, Inc., 2013.
3. Li, G., Z. Jin, M. Volpp, F. Otto, R. Lioutikov and G. Neumann, “ProDMP: A Unified Perspective on Dynamic and Probabilistic Movement Primitives”, *IEEE Robotics and Automation Letters*, Vol. 8, No. 4, pp. 2325–2332, 2023.
4. Calinon, S., “A Tutorial on Task-Parameterized Movement Learning and Retrieval”, *Intelligent Service Robotics*, Vol. 9, No. 1, pp. 1–29, January 2016.
5. Chi, C., Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake and S. Song, “Diffusion Policy: Visuomotor Policy Learning via Action Diffusion”, *The International Journal of Robotics Research*, 2024.
6. Brohan, A., N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, J. Hsu, J. Ibarz, B. Ichter, A. Irpan, T. Jackson, S. Jesmonth, N. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, K.-H. Lee, S. Levine, Y. Lu, U. Malla, D. Manjunath, I. Mordatch, O. Nachum, C. Parada, J. Peralta, E. Perez, K. Pertsch, J. Quiambao, K. Rao, M. Ryoo, G. Salazar, P. Sanketi, K. Sayed, J. Singh, S. Sontakke, A. Stone, C. Tan, H. Tran, V. Vanhoucke, S. Vega, Q. Vuong, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu and B. Zitkovich, “RT-1: Robotics Transformer for Real-World Control at Scale”, *arXiv preprint arXiv:2212.06817*, 2022.

7. Rezende, D. and S. Mohamed, “Variational Inference with Normalizing Flows”, F. Bach and D. Blei (Editors), *Proceedings of the 32nd International Conference on Machine Learning*, Vol. 37 of *Proceedings of Machine Learning Research*, pp. 1530–1538, PMLR, Lille, France, 07–09 Jul 2015.
8. Dinh, L., J. Sohl-Dickstein and S. Bengio, “Density estimation using Real NVP”, , 2017, <https://arxiv.org/abs/1605.08803>.
9. Zhang, J., H. B. Mohammadi and L. Rozo, “Learning Riemannian Stable Dynamical Systems via Diffeomorphisms”, K. Liu, D. Kulic and J. Ichnowski (Editors), *Proceedings of The 6th Conference on Robot Learning*, Vol. 205 of *Proceedings of Machine Learning Research*, pp. 1211–1221, PMLR, 14–18 Dec 2023.